

SPPU EXTERNAL EXAMINER

Viva & Practical Question Bank with Answers

Subject: Data Science and Big Data Analytics (DSBDA)

Third Year | Computer Engineering | 2019 Course | Savitribai Phule Pune University

15 Questions Per Unit | 10 Practical Questions | Total: 100 Q&A

UNIT I: Introduction to Data Science and Big Data

07 Hours | CO1

◆ Practical 1 & 2: Data Wrangling I & II

Q1. What is Data Science? Why is it important?

Ans: Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data. It is important because it enables data-driven decision-making, uncovers hidden patterns, supports predictive analytics, and drives business value.

Q2. What are the 5 V's of Big Data?

Ans: The 5 V's of Big Data are: (1) Volume – the sheer amount of data generated, (2) Velocity – the speed at which data is generated and processed, (3) Variety – different types of data (structured, semi-structured, unstructured), (4) Veracity – the quality and accuracy of data, and (5) Value – the usefulness derived from data.

Q3. What is the Data Science Life Cycle?

Ans: The Data Science Life Cycle includes: (1) Business Understanding, (2) Data Collection, (3) Data Preprocessing/Wrangling, (4) Exploratory Data Analysis, (5) Model Building, (6) Model Evaluation, and (7) Deployment and Monitoring.

Q4. What is data wrangling and why is it needed?

Ans: Data wrangling (also called data munging) is the process of cleaning, structuring, and enriching raw data into a desired format for better decision-making. It is needed because real-world data is often dirty, incomplete, inconsistent, or in unusable formats, which can affect model accuracy.

Q5. What is the difference between Business Intelligence and Data Science?

Ans: Business Intelligence (BI) focuses on descriptive analytics—analyzing historical data to understand what happened. Data Science goes beyond BI by using predictive and prescriptive analytics to forecast future events and recommend actions using machine learning and statistical modeling.

Q6. What are the different types of data?

Ans: Data types include: Structured data (tables/relational), Semi-structured data (JSON, XML), and Unstructured data (images, audio, text). Based on measurement: Nominal, Ordinal, Interval, and Ratio. Based on nature: Quantitative (numeric) and Qualitative (categorical).

Q7. What is data cleaning? Give examples of techniques.

Ans: Data cleaning is the process of detecting and correcting corrupt or inaccurate records from a dataset. Techniques include: handling missing values (imputation, deletion), removing duplicates, correcting inconsistencies, fixing structural errors, and filtering outliers.

Q8. What is data normalization and why is it used?

Ans: Data normalization is the process of scaling numerical features to a common range. It is used to prevent features with large values from dominating the model. Common methods are Min-Max Normalization (scales to [0,1]) and Z-score Standardization (mean=0, std=1).

Q9. What is data discretization?

Ans: Data discretization is the process of converting continuous numerical data into discrete intervals or categories (bins). For example, converting age values into groups like 0–18, 19–35, 36–60, 60+. It helps simplify models and reduce noise.

Q10. What is the relationship between Data Science and Information Science?

Ans: Information Science deals with the collection, classification, manipulation, storage, and dissemination of information. Data Science builds upon Information Science by adding statistical modeling, machine learning, and advanced analytics to extract actionable insights from data.

Q11. What is data integration?

Ans: Data integration involves combining data from multiple heterogeneous sources into a unified, coherent view. Techniques include data federation, data warehousing, and ETL (Extract, Transform, Load) processes.

Q12. What is data reduction? Name its techniques.

Ans: Data reduction aims to reduce the volume of data while preserving its integrity. Techniques include: Dimensionality Reduction (PCA), Numerosity Reduction (sampling, histograms), and Data Compression.

Q13. What is data transformation?

Ans: Data transformation converts data from one format or scale into another. Common transformations include: normalization, aggregation, generalization, attribute construction, and smoothing.

Q14. How do you handle missing values in a dataset?

Ans: Missing values can be handled by: (1) Deletion – remove rows/columns with missing values, (2) Mean/Median/Mode Imputation – replace with statistical measures, (3) Forward/Backward Fill, (4) Predictive Imputation using ML models, or (5) Using `isnull()` and `fillna()` in pandas.

Q15. What Python libraries are commonly used for data preprocessing?

Ans: Commonly used Python libraries: pandas (data manipulation), NumPy (numerical operations), Scikit-learn (preprocessing, modeling), Matplotlib and Seaborn (visualization), and SciPy (statistics).

UNIT II: Statistical Inference

07 Hours | CO2

◆ Practical 3: Descriptive Statistics – Measures of Central Tendency and Variability

Q1. Why is statistics important in Data Science?

Ans: Statistics provides the mathematical foundation for data analysis. It helps in summarizing data, identifying patterns, testing hypotheses, making predictions, and drawing conclusions from sample data about a population. Without statistics, data science models lack rigor and interpretability.

Q2. What are measures of central tendency? Name and explain them.

Ans: Measures of central tendency describe the center of a dataset: (1) Mean – arithmetic average of all values, (2) Median – middle value when data is sorted, (3) Mode – most frequently occurring value, (4) Mid-range – average of maximum and minimum values. Each is suitable for different types of distributions.

Q3. What is standard deviation and how does it differ from variance?

Ans: Variance measures the average squared deviation from the mean. Standard Deviation is the square root of variance, bringing it back to the original unit of measurement. Standard deviation is more interpretable as it is in the same unit as the data.

Q4. What is Bayes' Theorem?

Ans: Bayes' Theorem describes the probability of an event based on prior knowledge of related conditions. Formula: $P(A|B) = [P(B|A) \times P(A)] / P(B)$. It is widely used in Naïve Bayes classifiers, spam filters, and medical diagnosis.

Q5. What is hypothesis testing?

Ans: Hypothesis testing is a statistical method to make decisions about a population based on sample data. It involves: (1) Null Hypothesis (H_0) – assumes no effect, (2) Alternative Hypothesis (H_1) – assumes an effect, (3) Significance level (α), and (4) p-value comparison to decide whether to reject H_0 .

Q6. What is the p-value in hypothesis testing?

Ans: The p-value is the probability of observing results as extreme as the sample data, assuming H_0 is true. If $p\text{-value} < \alpha$ (typically 0.05), we reject H_0 . A small p-value indicates strong evidence against the null hypothesis.

Q7. What is Pearson Correlation Coefficient?

Ans: Pearson Correlation measures the linear relationship between two continuous variables. It ranges from -1 to +1: +1 = perfect positive correlation, -1 = perfect negative correlation, 0 = no linear correlation. Formula: $r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{(n \cdot \sigma_x \cdot \sigma_y)}$.

Q8. What is a Chi-Square test? When is it used?

Ans: The Chi-Square test is used to determine if there is a significant association between two categorical variables. It compares observed frequencies with expected frequencies. It is used in goodness-of-fit tests and tests of independence in contingency tables.

Q9. What is a t-test? What are its types?

Ans: A t-test is used to compare means between groups. Types: (1) One-sample t-test – compares sample mean to a known value, (2) Independent two-sample t-test – compares means of two independent groups, (3) Paired t-test – compares means of the same group at different times.

Q10. What is the difference between Mean Deviation and Standard Deviation?

Ans: Mean Deviation is the average of absolute deviations from the mean, treating all deviations equally. Standard Deviation gives more weight to larger deviations (due to squaring). Standard Deviation is more sensitive to outliers and is used more commonly in statistical inference.

Q11. What is the use of describe() function in pandas?

Ans: The describe() function generates summary statistics of a DataFrame including count, mean, standard deviation, min, 25th percentile (Q1), median (Q2), 75th percentile (Q3), and max for all numeric columns.

Q12. What is the IQR method for outlier detection?

Ans: The Interquartile Range (IQR) method detects outliers as values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. It is robust to extreme values and is used in box plots.

Q13. Explain Type I and Type II errors in hypothesis testing.

Ans: Type I Error (False Positive) – rejecting H_0 when it is actually true. Its probability is α (significance level). Type II Error (False Negative) – failing to reject H_0 when it is actually false. Its probability is β . The power of the test = $1 - \beta$.

Q14. What is Skewness and how does it affect data analysis?

Ans: Skewness measures the asymmetry of a distribution. Positive skew: tail on the right, mean > median. Negative skew: tail on the left, mean < median. Skewed data can violate assumptions of parametric tests and may require transformation.

Q15. How do you compute correlation in Python?

Ans: Using pandas: df.corr() computes pairwise Pearson correlation between all numeric columns. For specific variables: df['col1'].corr(df['col2']). For visualization, seaborn.heatmap(df.corr()) is commonly used.

UNIT III: Big Data Analytics Life Cycle

07 Hours | CO3

◆ Case Study: Global Innovation Social Network and Analysis (GINA)

Q1. What are the phases of the Big Data Analytics Life Cycle?

Ans: The 6 phases are: Phase 1: Discovery – understand business problem and data; Phase 2: Data Preparation – ETL, data warehouse setup; Phase 3: Model Planning – feature selection, algorithm selection; Phase 4: Model Building – training models; Phase 5: Communicate Results – present findings; Phase 6: Operationalize – deploy model in production.

Q2. What are the major sources of Big Data?

Ans: Sources include: Social media platforms (Twitter, Facebook), IoT sensors and devices, e-commerce transactions, healthcare records, financial transactions, log files, satellite imagery, and web scraping.

Q3. What is the Discovery phase in the analytics lifecycle?

Ans: In the Discovery phase, the data science team understands the problem domain, identifies key stakeholders, defines project goals, learns the business context, assesses available resources (data, tools, people), and frames initial hypotheses to be tested.

Q4. What happens in the Data Preparation phase?

Ans: Data Preparation involves: setting up the analytical sandbox, ETL (Extract, Transform, Load) processes, data exploration, data conditioning (cleaning, transforming), survey and visualize data, and establishing the data pipeline for model building.

Q5. What is the Model Planning phase?

Ans: In Model Planning, the team determines the methods and techniques to build relationships between variables, selects relevant variables, selects analytical techniques (regression, clustering, etc.), and creates a model framework based on tools like R, Python, or SAS.

Q6. What happens in the Communicate Results phase?

Ans: In this phase, the team presents findings to stakeholders, compares results against success criteria defined in the Discovery phase, identifies key findings and business value, chooses appropriate visualizations, and prepares reports and presentations.

Q7. What is the Operationalize phase?

Ans: In Operationalize, the final model is deployed into a production environment. It includes creating pilot projects, monitoring performance, automating pipelines, and integrating the model with existing business processes for real-time or batch predictions.

Q8. What is the GINA case study about?

Ans: GINA (Global Innovation Social Network and Analysis) is a case study demonstrating how Big Data analytics is applied to social networks. It involves analyzing social connections and patterns of innovators globally to identify influencers, collaboration networks, and innovation trends using graph analytics.

Q9. What is Big Data? How is it different from traditional data?

Ans: Big Data refers to datasets that are too large or complex for traditional data processing tools. Unlike traditional data (structured, smaller, manageable), Big Data is characterized by high Volume, Velocity, Variety, Veracity, and Value — requiring distributed processing systems like Hadoop and Spark.

Q10. What is the role of an analytical sandbox?

Ans: An analytical sandbox is a development environment where data scientists can explore, experiment, and build models freely without impacting production systems. It provides access to raw data and computational resources for discovery and modeling activities.

UNIT IV: Predictive Big Data Analytics with Python

07 Hours | CO4, CO2

◆ Practical 4, 5 & 6: Linear Regression, Logistic Regression, Naïve Bayes

Q1. What is Linear Regression? How does it work?

Ans: Linear Regression models the linear relationship between a dependent variable (Y) and one or more independent variables (X). The goal is to find the best-fit line: $Y = \beta_0 + \beta_1 X + \epsilon$. The coefficients are estimated using the Ordinary Least Squares (OLS) method to minimize the sum of squared residuals.

Q2. What is Logistic Regression? How is it different from Linear Regression?

Ans: Logistic Regression is used for binary classification. It models the probability of a class using the sigmoid function: $P(Y=1) = 1/(1+e^{(-z)})$. Unlike Linear Regression (continuous output), Logistic Regression outputs probabilities between 0 and 1 and predicts discrete classes.

Q3. What is the Naïve Bayes classifier?

Ans: Naïve Bayes is a probabilistic classifier based on Bayes' Theorem with the naïve assumption that all features are conditionally independent given the class label. Despite this simplification, it works well for text classification, spam filtering, and medical diagnosis.

Q4. What is a Decision Tree? Explain its structure.

Ans: A Decision Tree is a tree-shaped model where each internal node represents a feature, each branch represents a decision rule, and each leaf represents a class label or outcome. It splits data based on feature values using criteria like Gini Impurity or Information Gain (Entropy).

Q5. What is the Apriori Algorithm?

Ans: Apriori is a classic algorithm for finding frequent itemsets and generating association rules. It uses the anti-monotone property: if an itemset is infrequent, all its supersets are also infrequent. Key measures: Support, Confidence, and Lift.

Q6. What is the FP-Growth Algorithm? How is it better than Apriori?

Ans: FP-Growth (Frequent Pattern Growth) finds frequent itemsets without candidate generation using a compressed data structure called an FP-Tree. It is significantly faster than Apriori because it scans the dataset only twice and avoids multiple passes over the data.

Q7. What are Support, Confidence, and Lift in Association Rules?

Ans: Support = frequency of itemset in transactions. Confidence = $P(Y|X) = \text{Support}(X \cup Y) / \text{Support}(X)$, measuring rule reliability. Lift = Confidence/Support(Y), measuring how much more likely Y is given X compared to without X. Lift > 1 indicates a positive association.

Q8. What is the IRIS dataset?

Ans: The IRIS dataset contains 150 samples of iris flowers with 4 features: sepal length, sepal width, petal length, petal width, and 1 target variable (species: Setosa, Versicolor, Virginica). It is a benchmark dataset for classification in machine learning.

Q9. How do you handle missing values in Scikit-learn?

Ans: Using SimpleImputer: `from sklearn.impute import SimpleImputer`. Strategy options: 'mean', 'median', 'most_frequent', or 'constant'. The imputer is fit on training data and then transforms both train and test sets to prevent data leakage.

Q10. What is the difference between Predictive, Descriptive, and Prescriptive Analytics?

Ans: Descriptive Analytics: summarizes past data (what happened). Predictive Analytics: uses models to forecast future events (what will happen). Prescriptive Analytics: recommends actions to achieve desired outcomes (what should we do). Complexity increases from descriptive to prescriptive.

Q11. How do you remove duplicates in pandas?

Ans: Using `df.drop_duplicates()` to remove duplicate rows. `df.duplicated()` returns a boolean Series identifying duplicates. Parameters: subset (columns to consider), keep ('first', 'last', or False to drop all duplicates).

Q12. What is Label Encoding vs One-Hot Encoding?

Ans: Label Encoding assigns an integer to each category (e.g., 0, 1, 2). One-Hot Encoding creates binary columns for each category. Label Encoding is used for ordinal data; One-Hot Encoding for nominal data, as it avoids implying numerical order.

Q13. What is overfitting? How can it be prevented?

Ans: Overfitting occurs when a model learns the training data too well, including noise, leading to poor generalization. Prevention: cross-validation, regularization (L1/L2), pruning decision trees, reducing model complexity, and using more training data.

Q14. What is train-test split and why is it important?

Ans: Train-test split divides data into training (model learning) and testing (model evaluation) sets. Typical split: 80-20 or 70-30. It is important to evaluate model performance on unseen data and detect overfitting. In sklearn: `train_test_split()` from `sklearn.model_selection`.

Q15. What is Scikit-learn? Name some of its key modules.

Ans: Scikit-learn (sklearn) is a Python ML library built on NumPy, SciPy, and matplotlib. Key modules: `sklearn.preprocessing`, `sklearn.model_selection`, `sklearn.linear_model`, `sklearn.naive_bayes`, `sklearn.tree`, `sklearn.cluster`, `sklearn.metrics`, and `sklearn.datasets`.

UNIT V: Big Data Analytics and Model Evaluation

07 Hours | CO4, CO2

◆ Practical 7: Text Analytics | K-Means Clustering Evaluation

Q1. What is K-Means Clustering? Explain the algorithm.

Ans: K-Means partitions data into K clusters by minimizing within-cluster variance. Steps: (1) Randomly initialize K centroids, (2) Assign each point to nearest centroid, (3) Recalculate centroids as mean of assigned points, (4) Repeat until convergence. Uses Euclidean distance as default.

Q2. What is Hierarchical Clustering? Types?

Ans: Hierarchical Clustering builds a tree of clusters called a dendrogram. Types: (1) Agglomerative (bottom-up) – starts with each point as its own cluster and merges, (2) Divisive (top-down) – starts with all points in one cluster and splits. Linkage methods: single, complete, average, ward.

Q3. What is the Elbow Method in K-Means?

Ans: The Elbow Method determines the optimal number of clusters K by plotting the Within-Cluster Sum of Squares (WCSS) or inertia against K. The optimal K is where the plot shows an 'elbow' – a point of diminishing returns where adding more clusters doesn't significantly reduce WCSS.

Q4. What is a Confusion Matrix?

Ans: A Confusion Matrix is a table showing predicted vs actual class labels: TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative). From it, we compute: Accuracy = $(TP+TN)/(TP+TN+FP+FN)$, Precision = $TP/(TP+FP)$, Recall = $TP/(TP+FN)$, F1-Score = $2 \times P \times R / (P+R)$.

Q5. What is an ROC curve and AUC?

Ans: ROC (Receiver Operating Characteristic) curve plots True Positive Rate (Recall) vs False Positive Rate at various threshold values. AUC (Area Under Curve) measures overall classifier performance. AUC = 1.0 is perfect; AUC = 0.5 is random; AUC > 0.8 is considered good.

Q6. What is Time Series Analysis?

Ans: Time Series Analysis involves analyzing data collected at regular time intervals to identify trends, seasonality, and patterns. Methods include Moving Average, ARIMA, and Exponential Smoothing. Applications: stock forecasting, weather prediction, demand forecasting.

Q7. What is TF-IDF?

Ans: TF-IDF (Term Frequency-Inverse Document Frequency) measures the importance of a word in a document relative to a corpus. $TF = (\text{word count in doc})/(\text{total words in doc})$, $IDF = \log(\text{total docs} / \text{docs containing word})$. High TF-IDF means the word is important and unique to that document.

Q8. What is the Bag of Words model?

Ans: Bag of Words (BoW) represents text as a collection of word frequencies, ignoring grammar and word order. Each document is represented as a vector of word counts. It is simple but doesn't capture semantic meaning or word order.

Q9. What is text preprocessing? Name the steps.

Ans: Text preprocessing steps: (1) Tokenization – splitting text into tokens, (2) Stop word removal – removing common words (the, is, a), (3) Stemming – reducing words to root form (running→run), (4) Lemmatization – reducing to dictionary form, (5) Lowercasing, (6) Removing punctuation and special characters.

Q10. What is Social Network Analysis (SNA)?

Ans: Social Network Analysis studies relationships and interactions between entities (nodes) connected by links (edges). Key metrics: degree centrality, betweenness centrality, closeness centrality, and clustering coefficient. Tools: NetworkX in Python, Gephi.

Q11. What is the Holdout Method in model evaluation?

Ans: The Holdout Method reserves a portion of data as the test set and trains the model on the remaining data. Simple but results can vary based on how the split is made. Limitation: vulnerable to data partitioning bias.

Q12. What is Cross-Validation?

Ans: K-Fold Cross-Validation splits data into K equal parts, trains on K-1 folds, and tests on the remaining fold, repeating K times. The final score is the average. It provides a more reliable performance estimate than a single train-test split.

Q13. What is precision and how is it different from recall?

Ans: Precision = $TP / (TP + FP)$: of all positive predictions, how many are correct. Recall (Sensitivity) = $TP / (TP + FN)$: of all actual positives, how many were found. Precision focuses on prediction correctness; Recall focuses on completeness. F1-Score balances both.

Q14. What is Parameter Tuning? Name methods.

Ans: Parameter tuning finds optimal hyperparameters for a model. Methods: (1) Grid Search (GridSearchCV) – exhaustive search over parameter grid, (2) Random Search (RandomizedSearchCV) – random combinations, (3) Bayesian Optimization – intelligent search using probability models.

Q15. What are the limitations of K-Means clustering?

Ans: Limitations: (1) Requires pre-specifying K, (2) Sensitive to initial centroid placement, (3) Assumes spherical clusters of equal size, (4) Sensitive to outliers, (5) Struggles with non-convex cluster shapes. Solutions: K-Means++, multiple restarts, or using DBSCAN for arbitrary shapes.

UNIT VI: Data Visualization and Hadoop

07 Hours | CO5, CO6

◆ Practical 8, 9 & 10: Data Visualization using Seaborn & Matplotlib

Q1. What is Data Visualization? Why is it important?

Ans: Data Visualization is the graphical representation of data to communicate insights visually. It is important because humans process visual information faster than text, helps identify patterns, trends, and outliers, supports storytelling with data, and facilitates data-driven decision-making.

Q2. What are the types of data visualization?

Ans: Types include: Comparison charts (bar, column), Distribution charts (histogram, box plot), Relationship charts (scatter plot, bubble chart), Composition charts (pie chart, stacked bar), Trend charts (line plot), and Geographic charts (choropleth maps).

Q3. What is a Histogram? When is it used?

Ans: A Histogram shows the frequency distribution of continuous data divided into bins. It is used to understand data distribution, identify skewness, detect outliers, and check normality. X-axis shows value ranges; Y-axis shows frequency. In Python: `matplotlib.pyplot.hist()` or `seaborn.histplot()`.

Q4. What is a Box Plot? What information does it convey?

Ans: A Box Plot (box-and-whisker plot) shows the five-number summary: minimum, Q1 (25th percentile), median (Q2), Q3 (75th percentile), and maximum. Points beyond the whiskers are outliers. It is useful for comparing distributions and identifying outliers across groups.

Q5. What is a Scatter Plot? What is it used for?

Ans: A Scatter Plot displays the relationship between two continuous variables using points. It is used to identify correlation, detect clusters, and spot outliers. Color or size of points can represent a third variable. In Python: `matplotlib.pyplot.scatter()` or `seaborn.scatterplot()`.

Q6. What is Hadoop? What are its core components?

Ans: Hadoop is an open-source framework for distributed storage and processing of large datasets. Core components: (1) HDFS (Hadoop Distributed File System) – distributed storage, (2) MapReduce – distributed processing, (3) YARN – resource management. Part of the broader Hadoop ecosystem.

Q7. What is MapReduce?

Ans: MapReduce is a programming model for processing large datasets in parallel across a cluster. It has two phases: (1) Map – processes input data and produces key-value pairs, (2) Reduce – aggregates the intermediate key-value pairs to produce final output. Example: word count program.

Q8. What is Hive in the Hadoop ecosystem?

Ans: Apache Hive is a data warehouse tool built on top of Hadoop that provides SQL-like queries (HiveQL) for analyzing large datasets stored in HDFS. It translates HiveQL queries into MapReduce, Tez, or Spark jobs. Suitable for structured data analytics.

Q9. What is Apache Pig?

Ans: Apache Pig is a high-level platform for creating MapReduce programs using a scripting language called Pig Latin. It abstracts the complexity of MapReduce and is used for ETL tasks, data processing pipelines, and transformations on large datasets.

Q10. What are the challenges of Big Data Visualization?

Ans: Challenges include: (1) Scalability – visualizing millions of data points, (2) High dimensionality – displaying many variables, (3) Speed – real-time rendering, (4) Noise – distinguishing signal from noise, (5) User comprehension – making complex data understandable.

Q11. What is a Density Plot? How is it different from a Histogram?

Ans: A Density Plot (KDE – Kernel Density Estimate) shows the continuous probability distribution of a variable as a smooth curve. Unlike histograms (which depend on bin size), density plots provide a smoother representation. In seaborn: `sns.kdeplot()` or `sns.displot(kind='kde')`.

Q12. What is Seaborn? How is it different from Matplotlib?

Ans: Seaborn is a high-level statistical visualization library built on Matplotlib. It provides attractive default styles, built-in statistical functions, and simpler syntax for complex plots like heatmaps, violin plots, and pair plots. Matplotlib offers more customization at a lower level.

Q13. What is a Line Plot? When is it used?

Ans: A Line Plot displays data points connected by lines, showing trends over time or ordered categories. It is ideal for time series data. In Python: `matplotlib.pyplot.plot()` or `seaborn.lineplot()`.

Q14. What is the Titanic dataset and what insights can be drawn from it?

Ans: The Titanic dataset contains information on 891 passengers including age, sex, class, fare, and survival status. Key insights: survival rate was higher for women and children, first-class passengers had higher survival rates, and fare distribution was highly right-skewed.

Q15. What tools are used for Big Data Visualization?

Ans: Tools include: Tableau (interactive dashboards), Power BI (business intelligence), Apache Superset (open-source), D3.js (web-based), Python libraries (Matplotlib, Seaborn, Plotly, Bokeh), and Kibana (for Elasticsearch data).

UNIT ★: Practical Viva – Experiment-Specific Questions

Hours | All COs

◆ Direct Questions on Practicals 1 to 10

Q1. Practical 1: What does `isnull().sum()` do in pandas?

Ans: `isnull()` returns a DataFrame of boolean values (True for missing/NaN, False otherwise). `.sum()` counts True values per column, giving the total count of missing values in each column. This helps identify columns needing imputation or deletion.

Q2. Practical 2: What is an outlier? How do you detect it in Python?

Ans: An outlier is a data point significantly different from others. Detection methods: (1) IQR method – values $< Q1 - 1.5 \times IQR$ or $> Q3 + 1.5 \times IQR$, (2) Z-score method – $|z| > 3$ are outliers, (3) Box plots visually. In Python: `scipy.stats.zscore()` or manual IQR calculation using `df.quantile()`.

Q3. Practical 3: How do you calculate mean, median, mode in Python?

Ans: Using pandas: `df['col'].mean()` for mean, `df['col'].median()` for median, `df['col'].mode()[0]` for mode. Using NumPy: `np.mean()`, `np.median()`. For grouped stats: `df.groupby('category')['value'].describe()`.

Q4. Practical 4: How do you build a Linear Regression model in Scikit-learn?

Ans: Steps: (1) Import: `from sklearn.linear_model import LinearRegression`, (2) Split data: `train_test_split(X, y)`, (3) Create model: `model = LinearRegression()`, (4) Train: `model.fit(X_train, y_train)`, (5) Predict: `y_pred = model.predict(X_test)`, (6) Evaluate: `mean_squared_error(y_test, y_pred)`.

Q5. Practical 5: How do you compute a Confusion Matrix in Python?

Ans: `from sklearn.metrics import confusion_matrix, classification_report`. `cm = confusion_matrix(y_test, y_pred)`. For visualization: `seaborn.heatmap(cm, annot=True)`. `classification_report()` gives precision, recall, F1-score, and support for all classes.

Q6. Practical 6: How do you implement Naïve Bayes in Scikit-learn?

Ans: `from sklearn.naive_bayes import GaussianNB`. `model = GaussianNB()`. `model.fit(X_train, y_train)`. `y_pred = model.predict(X_test)`. `GaussianNB` is used for continuous features; `MultinomialNB` for text classification with word counts.

Q7. Practical 7: What is Tokenization?

Ans: Tokenization is the process of splitting text into smaller units (tokens), which can be words, sentences, or subwords. Word tokenization: 'Hello World' → ['Hello', 'World']. In Python: `word_tokenize()` from `nltk`, or `split()` for simple cases.

Q8. Practical 7: What is Stemming vs Lemmatization?

Ans: Stemming removes word suffixes to get the root form (e.g., 'running' → 'run', 'better' → 'better'). It is faster but may produce non-real words. Lemmatization uses vocabulary and morphological analysis to get the actual base form (e.g., 'better' → 'good'). Lemmatization is more accurate.

Q9. Practical 8 & 9: Why use Seaborn for the Titanic dataset?

Ans: Seaborn provides built-in support for statistical visualization with attractive default themes and easy integration with pandas DataFrames. It simplifies creating histograms, box plots, and heatmaps with just a few lines of code, and handles grouping and aggregation automatically.

Q10. Practical 10: How do you create a pairplot in Seaborn?

Ans: `seaborn.pairplot(df, hue='species')` creates a grid of scatter plots for all feature pairs, with diagonal plots showing distribution of each feature. The `hue` parameter colors points by class, making it easy to see class separability.

